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Use of Small Streams by Young Brook Trout Spawned in a Lake

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Abstract.—Young-of-the-year (age-0) brook trout *Salvelinus fontinalis* that were spawned in a lake were observed migrating into and inhabiting small (<2-m-wide) inlet streams. We estimated that up to 81% of the entire age-0 population in the lake system inhabited the streams by midsummer. Some age-0 fish overwintered in the streams and remained there during their second summer. Such behavior suggested fitness advantages were gained by stream residence. The inhabited streams had accessible, cold, and stable habitats that existed because the streams occurred in forested catchments where groundwater dominated base flows. This identification of these small streams as habitat for young brook trout and the importance of catchment hydrology for sustaining these habitats indicated a need to review habitat protection guidelines for lake-dwelling populations.

Brook trout *Salvelinus fontinalis* inhabit both lotic and lentic habitats throughout their range, and migrations between these habitats at different life history stages are common (Power 1980). Migrations from lentic to lotic habitats are typically associated with spawning (O'Conner and Power 1973; Keefe and Winn 1991); opposite migrations appear to be controlled by growth limitations (Saunders and Power 1970) and are common for young salmonids (e.g., Northcote 1969; Jonsson 1989). The migration of young salmonids spawned in lakes into tributary streams has generally been assumed to be incidental straying (Saunders and Power 1970; Jonsson 1989). However, Curry et al. (1993) observed young-of-the-year (age-0) brook trout migrating into and inhabiting small inlet streams of a lake.

A survey of lake-spawning populations of brook trout indicated that stream migration by age-0 fish may be a common behavior (Algonquin Fisheries Assessment Unit, Ontario Ministry of Natural Resources, unpublished data). The reasons for stream residence by these lake-spawned age-0 fish are uncertain. No apparent growth or feeding advantages are offered by the stream habitats during the initial 12 weeks after emergence (Curry et al. 1993). Nonetheless, small tributary streams can have greater relative abundance of organic drift and better growth rates for young salmonids than larger systems (Naiman et al. 1987; Erkinaro and Niemelä 1995). The physical and chemical differences between lake and stream environments may also influence growth and survival of age-0 fish possibly via differential rates of competition and predation. Stream migrations may be a function of genetic differences among trout within the lake system (e.g., Swain and Holtby 1989). These possibilities suggest that stream residence has significant implications for the evolutionary fitness of lake-dwelling populations and fisheries management of brook trout in lake systems. As a first assessment, we examined the spatial and temporal use of tributary streams by young trout spawned in a lake.

Methods

Meach Lake is located on the Precambrian Shield in central Ontario (45°18′N, 78°20′W; Figure 1). It is a relatively dilute, oligotrophic lake with specific conductance levels of 40 μS/cm. It has two basins, which have areas of 42 and 35 ha, and maximum depths of 11 and 26 m, respectively. Two shoreline spawning and incubation sites exist having a total area of 5 m^2. The watershed (670 ha) is dominated by a mixed hardwood forest. Eight streams discharge to the lake; there is one outlet stream at the north end. Only four of the

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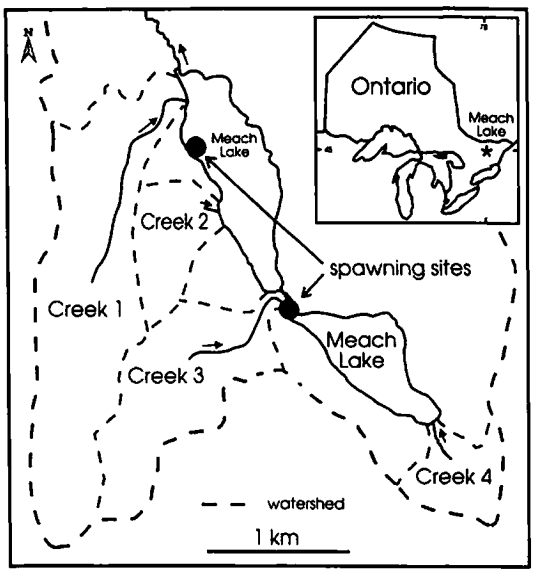

FIGURE 1.—Location of Meach Lake and the four tributary streams (creeks 1–4) used by young brook trout.

inlet streams are used by brook trout (creeks 1–4; Figure 1; Table 1). No spawning has been recorded in any stream.

The lake originally supported natural brook trout, but stocking of Hills Lake and Nipigon strains occurred annually from 1961–1986. The Nipigon strain originates from a lake-spawning population; the Hills Lake strain is mixed lake and riverine stocks. Stocking has since ceased, and the continued genetic contribution of stocked trout is unknown but assumed to be negligible (Danzmann et al. 1991). The lake also sustains a summer and winter fishery (approximately 500 trout averaging 25 cm in fork length harvested annually; C. Brady, unpublished data).

In 1992 and 1993, weirs were installed across the mouth of each stream. Each weir consisted of upstream, and downstream, staggered traps placed on opposite banks and connected by netting such that all migrants were captured. The weirs were installed and sampled daily for seven 5-d periods (installed May 15, June 29, and August 11, 1992, and June 15, July 12, August 10, and September 7, 1993). Numbers of migrating brook trout and total length (TL, mm) of each were recorded. All fish were then returned to the stream to continue their selected direction of travel.

The age-0 population in each stream was sampled during the periods of weir operation. In each stream, 10-m sections were isolated throughout the entire reach where age-0 fish could be encountered. They were captured by two fishery workers using dip nets for a period of 10–15 min and were held in large containers until the section sampling was complete. Total lengths (nearest mm) were recorded, and brook trout were returned to the section of capture.

TABLE 1.—Some physical characteristics of four study creeks and maximum distance and elevation of ascent above lake level by age-0 brook trout at Meach Lake, Ontario.

					Age-0 brook trout	
Creek	Width (cm)	Depth (cm)	Mid-channel velocity (cm/s)[a]	Total length (m)	Distance (m) from lake	Ascent (m) above lake
1	50–300	5–50	22	2,100	300	3.7
2	20–50	5–20	31	200	100	0.9
3	50–250	2–30	26	810	240	6.9
4	40–140	1–20	14	1,050	120	3.2

[a] Recorded 85–100 m upstream of the lake during June 1–2, 1995.

Two complete stream samples were collected during on 5-d period except during high-water conditions. On these occasions, the numbers of age-0 fish in unsampled 10-m sections were estimated from observed proportions in the same sections in previous or subsequent sampling periods relative to the entire stream. The two sampling sessions were used to estimate the total population by using an adjusted Petersen mark (upper caudal fin clip)–recapture model (Ricker 1975:78). After 3 years of study we assumed our dipnetting efficiency was consistent among streams, months, and years. The 1994 mark–recapture model provided the best estimates of the populations (greatest recapture:mark ratios); therefore, the 1994 total estimates were regressed on the average total number of age-0 fish captured during the two dipnetting sessions (estimated population = 2.49 × average total dipnetted; $r^2 = 0.94$; $N = 12$; $P < 0.01$), and this model was applied to the dipnetted values in 1992 and 1993 to give the reported population estimates in these years. Confidence limits were calculated from regression statistics (Sokal and Rohlf 1981).

The total number of age-0 brook trout in the entire Meach Lake system was estimated by applying a log–log decay curve between the mean total age-0 trout production for the lake and mean numbers of age-1 trout: $\log_{10}(\text{total age-0}) = -1.777 \times \log_{10}(\text{month}) + 4.138$. Total production of age-0 fish was estimated from six and eight traps (0.13 m^2) placed over the well-defined incubation

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habitats in 1992 and 1994, respectively (time = 0; see Curry et al. 1995). The number of alevins that emerged per trap averaged 69 (range, 3–182) in 1992 and 68 (5–138) in 1994. Extrapolating to the entire spawning and incubation habitats (5 m^2), total alevin production was estimated to be 13,858 in 1992 and 13,622 in 1994. The mean number of age-1 trout was obtained from annual assessments of the trout population from mark (fall trapnetting) and recapture (winter creel) studies during 1987–1994 (mean = 428 age-1 fish, range = 262–748, N = 6, and time = 18 months; C. Brady, unpublished data).

Total lengths were recorded for at least 10 age-0 fish per stream section. An overall mean of 87 fish per stream per month were measured (range, 2–197). Total lengths of age-0 fish were compared statistically among streams using analysis of variance controlling for sampling month and year (α = 0.01; SAS Institute 1989; Sokal and Rohlf 1981). Residuals and variances were examined and a logarithmic transformation was applied.

In 1993, samples of 10 age-0 fish from each stream and from the littoral zone of the lake were taken for mitochondrial DNA (mtDNA) analysis. Lake samples were taken from the littoral zone by using small mesh (5-mm, stretch diagonal measure) trap nets on July 20–30 and August 14–18. Nets were set farther than 100 m from stream mouths. Additional samples of 10 age-0 fish from creek 1 and 8 from the littoral zone taken in 1992 were included to increase sample sizes. Not all samples yielded DNA suitable for analysis.

Protocols for mtDNA extractions followed Chapman and Powers (1984) with the additional modification that 0.25 M sucrose was added to their TEK buffer for homogenization. Fragments of DNA were separated on either 0.8% or 1.0% agarose gels and transferred to nylon membranes (Hybond-N) with a Pharmacia-LKB vacublotter (30 min denaturation; 30 min neutralization; and 1.15-h transfer in 20 × standard sodium citrate). After overnight baking at 80°C, DNA fragments on the membranes were detected with DIG-labeled probes by following the method of Holtke et al. (1992). Probes were prepared from purified brook trout DNA fragments resulting from *Hind* III endonuclease digestion. The 11 restriction enzymes that generate restriction fragment length polymorphisms in brook trout (Danzmann et al. 1991) were used to analyze all fish. The resulting haplotypes obtained were classified according to Bernatchez and Danzmann (1993). Because sample sizes were small in the genetic analysis, differences among samples were tested by using a chi-square contingency test (χ^2) with Yalte's correction for continuity and a Monte Carlo bootstrapping algorithm.

TABLE 2.—Temperatures and depths of creeks 2 and 4 (C2 and C4) and Meach Lake, recorded at fixed locations during spring and summer, 1992–1994.

Month and measurement	1992			1993			1994		
	C2	C4	Lake	C2	C4	Lake	C2	C4	Lake
May									
Temperature (°C)	9	8	14						11
Depth (cm)	16	9	16						31
Jun									
Temperature (°C)				10	13	20			16
Depth (cm)				25	8	34			29
Jul									
Temperature (°C)	10	12	16	12	14	21	14	12	
Depth (cm)	12	5	17	28	10	34	17	6	
Aug									
Temperature (°C)	11	11	19	13	14	22	12	12	21
Depth (cm)	4	6	13	25	6	33	26	7	24
Sep									
Temperature (°C)				12	14	20	12	12	19
Depth (cm)				29	9	37	29	8	26

Differentiation of age-0 from yearling brook trout was based on TL validated by the recapture in 1992 of yearlings marked (removal of adipose fin) as age-0 fish in creeks 1 and 4 in 1991. Otoliths were also read, and no brook trout older than age 1 were encountered.

Stream and lake depths (fixed markers) and temperatures (hand-held thermometers) were measured at fixed locations near the stream mouths and at the spawning site in the lake's south basin (water depth < 1 m) between 1000 and 1400 hours. Stream depths were used as a surrogate for discharge (Gordon et al. 1992). Temperatures and depths in creeks 1 and 3 were found to be influenced by lake conditions; therefore creeks 2 and 4 are reported for lake–stream comparisons. Stream widths, depths, piezometric differences (elevation above lake level; Gordon et al. 1992), and temperature were measured at 10-m intervals during June 1–2, 1995. Flow rates were also measured at two sites 85–100 m upstream of the lake during this period (Gurley Pygmy current meter). Total stream length and maximum elevation above the lake were determined from the 1:10,000 Ontario base map series.

Results

Stream temperatures were consistently colder than the littoral zone of the lake (Table 2). Tem-

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TABLE 3.—Monthly mean population estimates of age-0 brook trout in four creeks (95% confidence intervals are in parentheses) and the total predicted population in the entire Meach Lake system, 1992–1994.

Year and month	Population size in creek: 1	2	3	4	Creek total	Total predicted age-0 trout	Stream resident (%)
1992							
May	87 (26–148)	102 (43–161)	511 (461–561)	2 (0–77)	702	6,074	12
Jun						3,768	
Jul	504 (455–553)	179 (131–227)	577 (518–636)	219 (175–263)	1,479	2,685	55
Aug	191 (145–238)	113 (56–170)	574 (516–633)	66 (2–130)	944	2,065	46
Sep						1,666	
1993							
May						6,074	
Jun	924 (807–1,041)	351 (313–388)	1,093 (945–1,241)	399 (360–438)	2,767	3,768	73
Jul	681 (606–756)	343 (254–331)	596 (534–657)	218 (173–262)	1,838	2,685	68
Aug	274 (234–313)	129 (74–184)	26 (0–97)	70 (0–126)	499	2,065	24
Sep	141 (87–194)	90 (29–150)	17 (0–89)	62 (0–127)	310	1,666	19
1994							
May						6,074	
Jun						3,768	
Jul	785 (655–952)	280 (152–414)	642 (453–733)	480 (274–541)	2,187	2,685	81
Aug	457 (279–547)	144 (40–289)	295 (311–581)	191 (78–338)	1,087	2,065	53
Sep	410 (220–486)	119 (0–244)	86 (0–229)	193 (93–353)	808	1,666	48

peratures were also consistent throughout each stream (creek 1 = 13.2 ± 0.4°C, n = 28; creek 2 = 9.9 ± 0.3°C, n = 10; creek 3 = 14.4 ± 0.5°C, n = 19; creek 4 = 11.0 ± 0.0°C, n = 15; mean ±SD; lake = 18°C; June 1–2, 1995).

Discharge regimes (i.e., stream depths) displayed seasonal trends (Table 2). Creeks 1 and 4 flowed continuously during the study. In creek 2, variable areas 60–100 m upstream flowed through underground channels after mid-June in each year. Creek 3 became dry upstream of 20 m except for several isolated pools (80–140 m) in 1993. Flow was reduced to less than 1-cm depths upstream of 20 m in creek 3 during 1994.

No age-0 fish were observed in the streams prior to May 1 in any year, but immigration into the streams was occurring by May 15 (1992 observations). Daily emigration and immigration among streams varied from 0 to 72 and from 0 to 107 age-0 fish/d, respectively. Yearlings were observed in the streams, but they were not captured in weirs in 1992. They were occasionally captured moving into creeks 2 and 4 from June to August in 1993 (total observed = 7). The only other species entering the streams was white sucker *Catostomus commersoni* using creek 1 for spawning and incubation.

The dispersal of age-0 fish upstream increased with time postemergence and varied annually according to habitat accessibility in each stream. Age-0 fish were consistently within 20 m of the lake by mid-May. They could be found throughout accessible regions of all streams by June in all years. The maximum upstream migration was determined by unsurpassable rapids or waterfalls. The maximum ascent gradient was 3% in creek 4.

Numbers of stream-resident age-0 brook trout varied among streams, months, and years (Table 3). The maximum numbers were observed in June and July. The midsummer maximum represented up to 81% of the estimated age-0 population in the lake system.

Age-0 fish increased in TL during the summer (Figure 2; September > August > July = June > May; $P < 0.05$). The largest age-0 brook trout were typically observed in creeks 1 and 4 (creek 1 = creek 4 > creek 2 > creek 3; $P < 0.05$). Smaller

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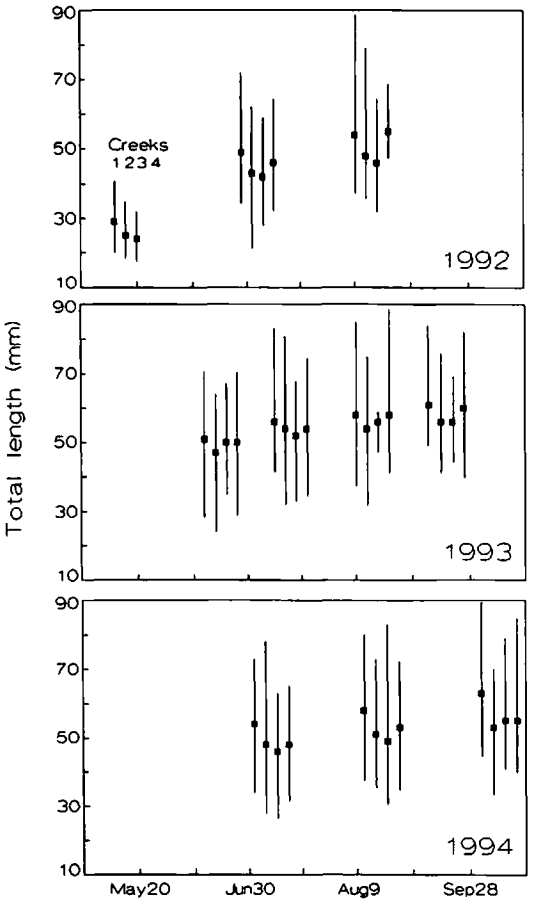

FIGURE 2.—Total lengths of age-0 brook trout in four Meach Lake tributary creeks in 1992, 1993, and 1994. Squares are means; vertical lines are ranges.

age-0 fish were observed in 1992 (1993 = 1994 > 1992; $P < 0.01$).

Yearlings were observed all years in each stream except during 1992 when none were observed in creek 3. Stream emigration and immigration by yearlings was less than 10 fish for the summer in any stream. Total observed ranged from 0 to 22 yearlings in any month, but 61 were captured with electrofishing gear in creek 1 during May 1992. In 1994, single yearlings were captured during September in creek 1 and during October in creek 3.

No significant differences in haplotype frequencies were observed between any of the stream populations or between the stream and lake age-0 brook trout (all pairwise Monte Carlo bootstrapping probabilities were >0.50; Table 4).

TABLE 4.—Observed distributions of mtDNA haplotypes among age-0 brook trout sampled in Meach lake and its tributary creeks.

	Number of fish with haplotype:		
Location	1	2	6
Creek 1	1	6	7
Creek 2		2	5
Creek 3		2	4
Creek 4		4	3
Lake		4	4

Discussion

The small tributary streams represented an important component of the habitats used by young brook trout in Meach Lake. Immigration into streams by lake-spawned age-0 fish began before the last alevins had emerged in May (Curry et al. 1995), and by midsummer up to 81% of the estimated age-0 population in the lake system were inhabiting the streams. Both age-0 and yearling (age-1) brook trout were observed in the streams from May to October and during midwinter (C. Brady, unpublished data). This extensive and intensive use of streams and the increasing observations of age-0 fish using streams in other lake systems demonstrates another variation in life history tactics used by brook trout (see review by Power 1980), and it suggests that fitness advantages are gained by migrating from a lake into streams.

Advantages are most likely related to the cold and stable thermal environment offered by the streams. The temperature of the littoral zone in the lake increased toward lethal levels for young brook trout by June (McCormick et al. 1972). Age-0 fish typically move offshore to deeper and presumably colder environments as littoral temperatures increase during summer (Venne and Magnan 1995). Given the species' propensity for cold, flowing waters (Power 1980) and the extensive dispersion of these fish after emergence from the gravel (creek 4 was >1 km from the closest incubation site; Griswold 1967), migrations into these cold streams would be predicted as was observed for older Arctic char *Salvelinus alpinus* (Näslund 1992).

It is not clear what specific fitness advantages are gained by stream residence. Achieving a maximum size during the first growing season is paramount to maximizing winter survival in salmonids (e.g., Gardiner and Geddes 1980). Growth differences between the stream and lake habitats were not observed in early summer (Curry et al. 1993) but may exist after longer periods of time (Erkinaro and Niemelä 1995; Randall et al. 1995). Growth may be enhanced in streams as age-0 fish

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learn to defend the microhabitats where growth can be enhanced (Grant and Noakes 1987; McLaughlin et al. 1992). Such habitats are less defined and less defendable in lake environments. A prediction of enhanced growth in streams requires further testing. Improved survival, either related to growth or to other environmental factors (e.g., predation rates) can not be detected until one or more year-classes can be followed through a generation.

Because brook trout from lacustrine and riverine sources have been stocked in Meach Lake, it is possible that stream migrations are linked to potential genetic differences (e.g., Swain and Holtby 1989). Our genetic analyses indicated no differences between lake- and stream-resident age-0 fish or among stream-resident age-0 fish. Sample sizes were relatively small in these analyses; however, numbers were adequate to identify and characterize the trout in question given the extensive examinations of mtDNA in brook trout populations in this area (Danzmann and Ihssen 1995). Streams are also used by lake-spawned age-0 fish in Dickson Lake, central Ontario (45°47′N, 78°12′W), where there is no history of stocking (Algonquin Fisheries Assessment Unit, Ontario Ministry of Natural Resources, unpublished data).

The persistent cold environments and flow of the streams occurred because they were located in forested catchments and, therefore, were sustained by groundwater inputs (Wright et al. 1990). Several of the unused streams also flowed through forested catchments, but their mouths were blocked and prevented entrance by age-0 fish. Age-0 brook trout in the lake would still congregate in the outflows of these streams. An additional unused stream flowed through an open lowland marsh that elevated stream temperatures to lake levels. The occurrence of the smallest young of the year in the least stable stream (creek 3) may be further evidence of the overall importance of a stable thermal environment.

Although questions remain unanswered regarding the specific consequences of stream residence, the sustained fishery in Meach Lake together with the presence of 80% of the age-0 fish in streams suggests the potential importance of the streams for supporting exploitable populations of lake-dwelling brook trout. Various land use activities can have negative impacts on salmoinds inhabiting streams (Barton et al. 1985; Hartman and Scrivener 1990). Because the use of streams has only recently been reported for lake-spawned brook trout (Curry et al. 1993), stream habitats may have been overlooked when protection guidelines were established for lake-dwelling brook trout; in addition, groundwater is clearly important for these stream habitats. It is, therefore, necessary for fisheries managers to reexamine habitat protection policies to ensure adequate protection for streams used by young brook trout and for the groundwater resources associated with the streams.

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References

Barton, D. R., W. D. Taylor, and R. M. Biette. 1985. Dimensions of riparian buffer strips required to maintain trout habitat in southern Ontario streams. North American Journal of Fisheries Management 5:364–378.

Bernatchez, L., and R. G. Danzmann. 1993. Congruence in control-region sequence and restriction site variation in mitochondrial DNA of brook charr (*Salvelinus fontinalis* Mitchill). Molecular Biology and Evolution 10:1002–1014.

Chapman, R. W., and D. A. Powers. 1984. A method for the rapid isolation of mitochondrial DNA from fishes. University of Maryland, Maryland Sea Grant Program, Technical Report UM-SG-TS-84-05, College Park.

Curry, R. A., S. Allen, M. G. Fox, and G. E. Morgan. 1993. Growth and food of young-of-the-year brook charr, *Salvelinus fontinalis* in lake and creek environments. Environmental Biology of Fishes 37: 131–138.

Curry, R. A., D. L. G. Noakes, and G. E. Morgan. 1995. Groundwater and the incubation and emergence of brook charr (*Salvelinus fontinalis*). Canadian Journal of Fisheries and Aquatic Sciences 52:1741–1749.

Danzmann, R. G., and P. E. Ihssen. 1995. A phylogeographic survey of brook charr (*Salvelinus fontinalis*) in Algonquin Park, Ontario, based upon mitochondrial DNA variation. Molecular Ecology 4:681–697.

Danzmann, R. G., P. E. Ihssen, and P. D. N. Hebert. 1991. Genetic discrimination of wild and hatchery populations of brook charr, *Salvelinus fontinalis* (Mitchill), in Ontario using mitochondrial DNA analysis. Journal of Fish Biology 39(Supplement A):69–77.

Erkinaro, J., and E. Niemelä. 1995. Growth differences between the Atlantic salmon parr, *Salmo salar*, of nursery brooks and natal rivers in the River Teno

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watercourse in northern Finland. Environmental Biology of Fishes 42:277–287.

Gardiner, W. R., and P. Geddes. 1980. The influence of body composition on the survival of juvenile salmon. Hydrobiologia 69:67–72.

Gordon, N. D., T. A. McMahon, and B. L. Finlayson. 1992. Stream hydrology. An introduction for ecologists. Wiley, Toronto.

Grant, J. W. A., and D. L. G. Noakes. 1987. Movers and stayers: foraging tactics of young-of-the-year brook charr, *Salvelinus fontinalis*. Journal of Animal Ecology 56:1001–1013.

Griswold, B. L. 1967. Some aspects of the shore spawning of brook trout (*Salvelinus fontinalis* Mitchill). Master's thesis. University of Maine, Orono.

Hartman, G. F., and J. C. Scrivener. 1990. Impacts of forestry practices on a coastal stream ecosystem, Carnation Creek, British Columbia. Canadian Bulletin of Fisheries and Aquatic Sciences 223.

Holtke, H. J., G. Sanger, C. Kessler, and G. Schmitz. 1992. Sensitive chemiluminescent detection of digoxigenin-labelled nucleic acids: a fast and simple protocol and its applications. BioTechniques 12: 104–112.

Jonsson, B. 1989. Life history and habitat use of Norwegian brown trout (*Salmo trutta*). Freshwater Biology 21:71–86.

Keefe, M., and H. E. Winn. 1991. Chemosensory attraction to home stream water and conspecifics by native brook trout, *Salvelinus fontinalis*, from two southern New England streams. Canadian Journal of Fisheries and Aquatic Sciences 48:938–944.

McCormick, J. H., K. E. F. Hokanson, and B. R. Jones. 1972. Effects of temperature on growth and survival of young brook trout, *Salvelinus fontinalis*. Journal of the Fisheries Research Board of Canada 29:1107–1112.

McLaughlin, R. L., J. W. A. Grant, and D. L. Kramer. 1992. Individual variations and alternative patterns of foraging movements in recently-emerged brook charr. Behaviour 120:286–301.

Naiman, R. J., J. M. Melillo, M. A. Lock, T. E. Ford, and S. R. Reice. 1987. Longitudinal patterns of ecosystem processes and community structure in a subarctic river continuum. Ecology 68:1139–1156.

Näslund, I. 1992. Upstream migratory behaviour in landlocked Arctic charr. Environmental Biology of Fishes 33:265–274.

Northcote, T. G. 1969. Patterns and mechanisms in the lakeward migratory behaviour of juvenile trout. Pages 183–203 *in* T. G. Northcote, editor. Salmon and trout in streams. H. R. MacMillan Lectures in Fisheries, University of British Columbia, Vancouver.

O'Connor, J. F., and G. Power. 1973. Homing of brook trout (*Salvelinus fontinalis*) in Matamek Lake, Quebec. Journal of the Fisheries Research Board of Canada 30:1012–1014.

Power, G. 1980. The brook charr, *Salvelinus fontinalis*. Pages 141–203 *in* E. K. Balon editor. Charrs. Salmonid fishes of the genus *Salvelinus*. Dr W. Junk, The Hague, The Netherlands.

Randall, R. G., J. R. M. Kelso, and C. K. Minns. 1995. Fish production in freshwaters: Are rivers more productive than lakes? Canadian Journal of Fisheries and Aquatic Sciences 52:631–643.

Ricker, W. E. 1975. Computation and interpretation of biological statistics of fish populations. Fisheries Research Board of Canada Bulletin 191.

SAS Institute. 1989. SAS/STAT user's guide, version 6, 4th edition. SAS Institute, Cary, North Carolina.

Saunders, L. H., and G. Power. 1970. Population ecology of brook trout, *Salvelinus fontinalis*, in Matamek Lake, Quebec. Journal of the Fisheries Research Board of Canada 27:413–424.

Sokal, R. R., and F. J. Rohlf. 1981. Biometry, 2nd edition. Freeman, New York.

Swain, D. P., and L. B. Holtby. 1989. Differences in morphology and behaviour between juvenile coho salmon (*Oncorhynchus kisutch*) reared in a lake and its tributary stream. Canadian Journal of Fisheries and Aquatic Sciences 46:1406–1414.

Venne, H., and P. Magnan. 1995. The impact of intraspecific and interspecific interactions of young-of-the-year brook charr in temperate lakes. Journal of Fish Biology 46:669–686.

Wright, K. A., K. H. Sendek, R. M. Rice, and R. B. Thomas. 1990. Logging effects on streamflow: storm runoff at Casper Creek in northwestern California. Water Resources Research 26:1657–1667.

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